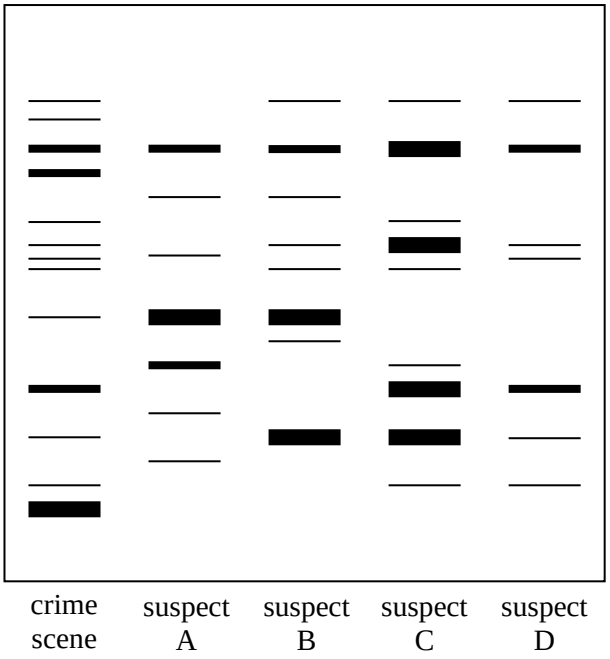
Multiple-choice questions

|!|BCEMB0280000001|!| ★

The diagram below shows the DNA fingerprints produced from samples collected at a crime scene as well as from four suspects.



Which suspect is ***most*** likely to have been at the crime scene?

A. suspect A
B. suspect B
C. suspect C
D. suspect D

□

##D##

Skill assessed	Mark
Analysing data	1

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|!|BCEMC0280000002|!|

Arrange in the correct order the following steps in the production of human insulin using GM bacteria.

(1) Insert the human insulin gene into bacterial plasmids.
(2) Culture the bacteria in large quantities and extract human insulin from the culture.
(3) Cut out the insulin gene from human DNA.
(4) Place the plasmids into bacteria.

A. (1) → (3) → (2) → (4)
B. (1) → (3) → (4) → (2)
C. (3) → (1) → (2) → (4)
D. (3) → (1) → (4) → (2)

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0280000003|!|

Which of the following is ***not*** an application of the Human Genome Project?

A. improving the diagnosis of diabetes mellitus
B. health risk assessment
C. finding the evolutionary relationships between humans and other organisms
D. cloning new individuals

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0280000004|!| ★

Which of the following statements about plasmids are ***incorrect***?

A. Plasmids are small circular DNA molecules found in bacteria.
B. Replication of the plasmid is synchronized with replication of the bacterial chromosome.
C. Plasmids carry genes that are not essential for survival.
D. Plasmids can be transferred between bacteria.

##B##

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Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0280000005|!|

Which of the following is/are the application(s) of recombinant DNA technology?

(1) producing human insulin using bacteria
(2) increasing the yield of crops
(3) identifying victims in disasters

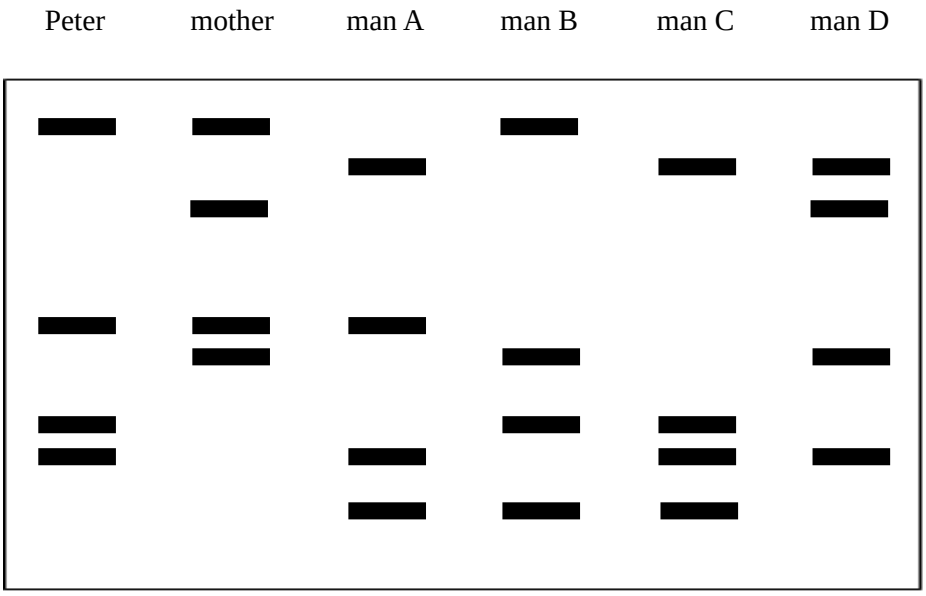
A. (2) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0280000006|!| ★

The following diagram shows the DNA fingerprints of Peter, his mother and four men.



Which man is ***most*** probably the biological father of Peter?

A. man A
B. man B
C. man C
D. man D

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0280000007|!|

Which of the following statement(s) about gel electrophoresis is/are correct?

(1) An electric field is used to drive DNA fragments across a gel.
(2) DNA fragments move from the positive end towards the negative end of the gel.
(3) The distance moved by a DNA fragment is proportional to the fragment size.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMA0280000008|!| ★★ **Cross Topic**

A human cell has between 20,000 and 25,000 genes while an *E. coli* cell has approximately 4000 genes. Which of the following conclusion(s) can be drawn?

(1) The genome of humans is larger than that of *E.coli*.
(2) Some genes are common to humans and *E.coli*.
(3) The human body can produce more types of proteins than *E.coli*.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

##B##

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Skill assessed	Mark
Applying concepts	1

|!|BCEMC0280000009|!|

Which of the following enzymes are involved in recombinant DNA technology?

(1) restriction enzyme
(2) DNA polymerase
(3) DNA ligase

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0280000010|!|

Which of the following is ***not*** an advantage of applying recombinant DNA technology in agriculture?

A. increased tolerance of crops to salt
B. increased resistance of crops to diseases
C. increased nutritional value of crops
D. increased dependence on chemical pesticides

##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0280000011|!| ★

Which of the following features of bacteria make them suitable for use in recombinant DNA technology?

(1) They have a fast growth rate.
(2) They can take up plasmids.
(3) Plasmids often carry genes that can be used for screening.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0280000012|!|

Which of the following is ***not*** an application of DNA fingerprinting?

A. parentage testing
B. screening for genetic diseases
C. sequencing of the human genome
D. authentication of Chinese medicines

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0280000013|!|

Which of the following statements about the use of gel electrophoresis to separate DNA fragments is/are correct?

(1) Smaller DNA fragments travel faster in the gel.
(2) DNA fragments move towards the negative end of the gel.
(3) Methylene blue solution may be used to stain DNA bands.

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

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##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0280000014|!| ★

DNA fingerprinting ***cannot*** be used to differentiate between

A. people with and without genetic defects.
B. clones.
C. fraternal twins.
D. ancient ancestors.

##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0280000015|!|

Which of the following is/are the aim(s) of the Human Genome Project?

(1) sequence human DNA
(2) make improvements to the human genome
(3) identify human genetic diseases

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0280000016|!|

Which of the following are the social concerns that were raised by the Human Genome Project?

(1) the right to use personal genetic information
(2) discrimination against people with certain genetic traits
(3) the unknown role of genes

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0280000017|!|

Which of the following is ***not*** an application of genetically modified bacteria?

A. production of vaccines
B. identification of victims in disasters
C. extraction of metals from metal ores
D. cleaning up of oil spills

##B##

Skill assessed	Mark
Understanding basic concepts	1

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